

UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO



FACULTAD DE INGENIERÍA

DIVISIÓN DE INGENIERÍA ELÉCTRICA

Temas selectos de Ing. En Computación II

GRUPO 02

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Tarea Ejercicios Ansible

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Evidencia de ejercicios ANSIBLE

01 - Paso a paso en la VM

1. Crear el usuario ansible y darle nombre al host

```
[root@localhost ~]# useradd ansible
Creando el fichero del buzón de correo: El fichero ya existe
[root@localhost ~]# passwd ansible
Cambiando la contraseña del usuario ansible.
Nueva contraseña:
Vuelva a escribir la nueva contraseña:
passwd: todos los tokens de autenticación se actualizaron exitosamente.
[root@localhost ~]#  echo "127.0.0.1   nodo1" >> /etc/hosts
[root@localhost ~]# getent hosts nodo1
127.0.0.1       nodo1
```

2. Conectarse a nodo1 (misma máquina) usando password

```
[root@localhost ~]# ssh ansible@nodo1
ansible@127.0.0.1's password:
Last failed login: Fri Oct 23 08:03:00 CST 2026 from 127.0.0.1 on ssh:notty
There were 2 failed login attempts since the last successful login.
[ansible@localhost ~]$ exit
cerrar sesión
Connection to 127.0.0.1 closed.
[root@localhost ~]# cat .ssh/known_hosts
nodo1 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGkauMvuOfkbKsDgkL7brzpEb0qCzHs/547r2IyWIx11
nodo1 ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABgQDXpIeOV8BBTdwHuwDAyPBr1ZoXWXI1tqyL2Sp08ZngA3u+eA1GpMoJjzqmGpK05
Z8lQgD10hvqYcb9Nu6zwqGY+9LaPiIZ8tPBzEI9djtciwK5aoZ82jZQcJB3X+Cp8H17B957GN5ebInA/4xLsENFZTAHOc
6ePbYuTtVcYWZoS/4Yp1jQi49k+Ueex/h3wJ9qiqPUxIPnwFzhJkJtNg2ZYiE+qloXidj1pcwLPiaDN8DmoW8IBDfAq/X
XWQY74UqRJ9+sGJ04OmEmwmQHE2cIyEl+goDBeuOPe3w4CoUw+I2ebxp2ANFv5xeIFGW3WOkl+zQsueyYH/0/tV3U23ra
gi+N8BAY7g2qGwijAKQj+rf5wX1Jfrl3eeEq+QaQkW/qnSMD5irjISTgPAItByrdwhxeqSBe9ENF0kgQHqqtd5cmQ9zaX
9PDPsCec6ivNgz3mo8ebI0pcY7pB42pq5bCXY+YT4O0NbQNzbQcuIqYiSFDTAPWHugfZwq3cSC0kwc=
nodo1 ecdsa-sha2-nistp256
AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBB1NnJN3pImt6mydrtDvK2LKmzCcYOBFPjhlP+CTY
xD3Wrlx2I+K6KDR5jTnYNR9Y7o+N+Dgh6moUQ1KNXeqTPQ=
127.0.0.1 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGkauMvuOfkbKsDgkL7brzpEb0qCzHs/547r2IyWIx11
192.168.122.40 ssh-ed25519
AAAAC3NzaC1lZDI1NTE5AAAAIGkauMvuOfkbKsDgkL7brzpEb0qCzHs/547r2IyWIx11
```

3. Generar claves ssh para el nodo de control (con usuario root

```
[root@localhost ~]# # ssh-keygen -t ed25519 -C "root@control"
[root@localhost ~]# ls -l ~/.ssh/
total 28
-rw-r--r--. 1 root root  109 oct 23 09:32 config
-rw-------. 1 root root  432 oct 23 09:17 github_mquinteror1000
-rw-------. 1 root root  113 oct 23 09:17 github_mquinteror1000.pub
-rw-------. 1 root root  419 oct 23 09:36 id_ed25519
-rw-r--r--. 1 root root  108 oct 23 09:36 id_ed25519.pub
-rw-------. 1 root root 1000 oct 23 09:17 known_hosts
-rw-r--r--. 1 root root   87 oct 23 08:16 known_hosts.old
[root@localhost ~]# cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIFVWx2Thz5aILUhyG2xb0OjQ9X8QXk5E1JdObixSIqcw
root@localhost.localdomain
[root@localhost ~]#
```

Se puede ver que si se generó el par de claves

4. usar ssh-coy-id para configurar acceso mediante clave ssh

```
[root@localhost ~]# ssh-copy-id ansible@nodo1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that
are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is
to install the new keys
ansible@127.0.0.1's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'ansible@nodo1'"
and check to make sure that only the key(s) you wanted were added.

[root@localhost ~]# ssh ansible@nodo1
Last login: Thu Oct 29 18:31:35 2026
```

Funciona!!

5. ejecutando comandos remotos

```
[root@localhost ~]# ssh ansible@nodo1 'hostname; whoami; uptime'
localhost.localdomain
ansible
 18:36:59 up  1:11,  1 user,  load average: 0.01, 0.03, 0.00
[root@localhost ~]# for h in nodo1 nodo1 nodo1; do ssh ansible@$h 'date'; done
jue 29 oct 2026 18:37:07 CST
jue 29 oct 2026 18:37:07 CST
jue 29 oct 2026 18:37:07 CST
[root@localhost ~]#
```

Han funcionado, como entra por ssh directo autenticando con la clave.

Provocar una falla de autenticacion a propósito, moficando los permisos

```
[root@localhost ~]# chmod 777 /home/ansible/.ssh
[root@localhost ~]# ssh ansible@nodo1
ansible@nodo1's password:
```

Ahora pide password, porque el directorio /home/ansible/.ssh no cumple con las politicas de permisos para ser usado

Diagnosticar con ssh modo verboso ssh -v y filtrando Autentications

```
[root@localhost ~]# ssh -v ansible@nodo1 2>&1 | grep -i -E "offering|authentications"
debug1: Authentications that can continue: publickey,gssapi-keyex,gssapi-with-mic,password
debug1: Offering public key: /root/.ssh/id_ed25519 ED25519
SHA256:9vWA7QZ5hgKorSlO01bHKcXcNzBb1JTvidXtRqWnpD4
ansible@nodo1's password: debug1: Authentications that can continue: publickey,gssapi-
keyex,gssapi-with-mic,password
```

sabemos no ha querido autenticar con claves ssh, por eso pide password.

Arreglemoslo, dejando los permisos como se remcomienda 700 para /home/ansible/.ssh y 600 para /home/ansible/.ssh/authorized_keys

```
[root@localhost ~]# chmod 700 /home/ansible/.ssh
[root@localhost ~]# chmod 600 /home/ansible/.ssh/authorized_keys
[root@localhost ~]# ssh ansible@nodo1 hostname
localhost.localdomain
[root@localhost ~]# ssh ansible@nodo1
Last login: Thu Oct 29 18:34:47 2026 from 127.0.0.1
```

ha funcionado, como lo hcemos todo como root, podemos usmear los home de otros usuarios, pero lo correcto sería que ansible sea quien corrija sus permisos

02 - copiar archivos de un punto a otro

1. scp: origen -> destino

local a remoto

```
[root@localhost ~]# echo "Reporte de aves - $(date)" > /root/reporte.txt
[root@localhost ~]# scp /root/reporte.txt ansible@nodo1:/home/ansible/
reporte.txt
```

remoto a local

```
[root@localhost ~]# scp ansible@nodo1:/etc/os-release /tmp/os-release-nodo1
os-release
```

carpeta completa

```
[root@localhost ~]# scp ansible@nodo1:/etc/os-release /tmp/os-release-nodo1
os-release
100%  579     1.0MB/s   00:00
[root@localhost ~]# mkdir -p /root/sitio && for i in 1 2 3; do echo "pagina $i" >
/root/sitio/p$i.html; done
[root@localhost ~]# scp -r /root/sitio ansible@nodo1:~
p2.html
100%    9    23.2KB/s   00:00
p3.html
100%    9    39.2KB/s   00:00
p4.html
100%    6    33.0KB/s   00:00
p1.html
100%    9    45.4KB/s   00:00
[root@localhost ~]# ssh ansible@nodo1 'ls -l ~/sitio ~/reporte.txt'
-rw-r--r--. 1 ansible ansible 47 oct 29 19:07 /home/ansible/reporte.txt

/home/ansible/sitio:
total 16
-rw-r--r--. 1 ansible ansible 9 oct 29 19:09 p1.html
-rw-r--r--. 1 ansible ansible 9 oct 29 19:09 p2.html
-rw-r--r--. 1 ansible ansible 9 oct 29 19:09 p3.html
-rw-r--r--. 1 ansible ansible 6 oct 29 19:09 p4.html
```

2. rsync solo lo que cambió

```
[root@localhost ~]# dnf install -y rsync
Última comprobación de caducidad de metadatos hecha hace 0:37:55, el jue 29 oct 2026
18:33:51.
El paquete rsync-3.2.7-1.el9_8.x86_64 ya está instalado.
Dependencias resueltas.
Nada por hacer.
¡Listo!
[root@localhost ~]#  rsync -avz /root/sitio/ ansible@nodo1:/home/ansible/respaldo/
sending incremental file list
created directory /home/ansible/respaldo
./
p1.html
p2.html
p3.html
p4.html

sent 349 bytes  received 140 bytes  326.00 bytes/sec
total size is 33  speedup is 0.07
[root@localhost ~]# rsync -avz /root/sitio/ ansible@nodo1:/home/ansible/respaldo/
sending incremental file list

sent 139 bytes  received 12 bytes  100.67 bytes/sec
total size is 33  speedup is 0.22
[root@localhost ~]# echo "nueva" > /root/sitio/p4.html
[root@localhost ~]# rsync -avz /root/sitio/ ansible@nodo1:/home/ansible/respaldo/
sending incremental file list
p4.html

sent 180 bytes  received 41 bytes  147.33 bytes/sec
total size is 33  speedup is 0.15
[root@localhost ~]# rm /root/sitio/p1.html
rm: ¿borrar el fichero regular '/root/sitio/p1.html'? (s/n) s
[root@localhost ~]# rsync -avz --delete --dry-run /root/sitio/
ansible@nodo1:/home/ansible/respaldo/
sending incremental file list
deleting p1.html
./

sent 124 bytes  received 33 bytes  314.00 bytes/sec
total size is 24  speedup is 0.15 (DRY RUN)
```

3. sftp modo interactivo

```
[root@localhost ~]# sftp ansible@nodo1
Connected to nodo1.
sftp> wd ; ls
Invalid command.
sftp> pwd
Remote working directory: /home/ansible
sftp> ld
Invalid command.
sftp> ls
reporte.txt  respaldo     sitio
sftp> get reporte.txt /tmp/reporte-bajado.txt
Fetching /home/ansible/reporte.txt to /tmp/reporte-bajado.txt
open local "/tmp/reporte-bajado.txt": Permission denied
sftp> put /etc/hostname
Uploading /etc/hostname to /home/ansible/hostname
hostname
100%    1     0.9KB/s   00:00
sftp> bye
```

4. Motivación para ansible

¿Por qué ansible es mejor que un script interactivo?

```
[root@localhost ~]# vim configura.sh
```

con este contendido

```
#!/bin/bash
for h in nodo1; do
  ssh ansible@$h 'echo "zona=aves" >> ~/config.txt'
done
```

se ejecuta 3 veces

```
[root@localhost ~]# sh configura.sh
[root@localhost ~]# sh configura.sh
[root@localhost ~]# sh configura.sh
[root@localhost ~]# ls
anaconda-ks.cfg  configura.sh  devops2026  evidencia-examen.sh  evidencia-martin-quintero.txt
reporte.txt  script-evidencia.sh  sitio
[root@localhost ~]# cat /home/ansible/config.txt
zona=aves
zona=aves
zona=aves
[root@localhost ~]#
```

efectivamente, tres veces hizo lo mismo

Parte 2 Ansible

1. Instalar paquetes

Debe ser **ansible-core** el paquete a instalar, no ansible u otro parecido, solo el base

```
[root@localhost ~]# dnf install -y ansible-core
Última comprobación de caducidad de metadatos hecha hace 0:53:39, el jue 29 oct 2026
18:33:51.
El paquete ansible-core-1:2.14.18-3.el9_8.1.x86_64 ya está instalado.
Dependencias resueltas.
Nada por hacer.
¡Listo!
[root@localhost ~]# ansible --version
ansible [core 2.14.18]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/root/.ansible/plugins/modules',
'/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.9.25 (main, Aug  6 2026, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-
14)] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
[root@localhost ~]# mkdir -p /root/devops2026/ansible && cd /root/devops2026/ansible
[root@localhost ansible]# pwd
/root/devops2026/ansible
```

Hasta aqui esta instalado el paquete y creado el directorio /root/devops2026/ansible

2. Archivos de configuracion en /root/devops2026/ansible

En el directorio de trabajo se crean los archivos **inventario.ini** y **ansible.cfg** cada uno con su respectivo contendido segun el manual del profesor

```
[root@localhost ansible]# touch inventario.ini
[root@localhost ansible]# touch ansible.cfg
[root@localhost ansible]# vim inventario.ini
[root@localhost ansible]# vim ansible.cfg
[root@localhost ansible]# cat inventario.ini
[web]
nodo1  ansible_host=127.0.0.1

[zoologico:children]
web
[root@localhost ansible]# car ansible.cfg
-bash: car: orden no encontrada
[root@localhost ansible]# cat ansible.cfg
[defaults]
inventory   = ./inventario.ini
remote_user = ansible
forks       = 5

[root@localhost ansible]#
```

verificar que esté correcta la configuración

```
[root@localhost ansible]# ansible --version | grep "config file"
  config file = /root/devops2026/ansible/ansible.cfg
[root@localhost ansible]# ansible-inventory --graph
@all:
  |--@ungrouped:
  |--@zoologico:
  |  |--@web:
  |  |  |--nodo1
[root@localhost ansible]# ansible-inventory --list
{
    "_meta": {
        "hostvars": {
            "nodo1": {
                "ansible_host": "127.0.0.1"
            }
        }
    },
    "all": {
        "children": [
            "ungrouped",
            "zoologico"
        ]
    },
    "web": {
        "hosts": [
            "nodo1"
        ]
    },
    "zoologico": {
        "children": [
            "web"
        ]
    }
}
[root@localhost ansible]# ansible all --list-hosts
  hosts (1):
    nodo1
[root@localhost ansible]#
```

Parace que está bien

2. Actividades Básicas (comandos ad-hoc)

sintaxis básica de un comando ad-hoc

```
ansible <hosts> -m <módulo> -a "<argumentos>" [-b]
```

1. Conectividad: prueba ssh y ping en el nodo

```
[root@localhost ansible]# ansible all -m ansible.builtin.ping
nodo1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
[root@localhost ansible]#
```

funcionó

2. Facts: Información que puede obtenera ansible sobre el nodo

```
[root@localhost ansible]# ansible nodo1 -m ansible.builtin.setup -a
"filter=ansible_distribution*"
nodo1 | SUCCESS => {
    "ansible_facts": {
        "ansible_distribution": "Rocky",
        "ansible_distribution_file_parsed": true,
        "ansible_distribution_file_path": "/etc/redhat-release",
        "ansible_distribution_file_variety": "RedHat",
        "ansible_distribution_major_version": "9",
        "ansible_distribution_release": "Blue Onyx",
        "ansible_distribution_version": "9.8",
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false
}
[root@localhost ansible]# ansible nodo1 -m ansible.builtin.setup -a
"filter=ansible_memtotal_mb"
nodo1 | SUCCESS => {
    "ansible_facts": {
        "ansible_memtotal_mb": 1707,
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false
}
[root@localhost ansible]# ansible nodo1 -m ansible.builtin.command -a "df -h /"
nodo1 | CHANGED | rc=0 >>
S.ficheros          Tamaño Usados  Disp Uso% Montado en
/dev/mapper/rl-root    16G   2.3G   14G  15% /
```

Valla

3. Instalar un paquete

Esta primera ha de fallar

```
[root@localhost ansible]# ansible web -m ansible.builtin.dnf -a "name=tree state=present"
nodo1 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "msg": "This command has to be run under the root user.",
    "results": []
}
[root@localhost ansible]# ansible web -b -m ansible.builtin.dnf -a "name=tree state=present"
nodo1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "msg": "Nothing to do",
    "rc": 0,
    "results": []
}
[root@localhost ansible]#
```

bueno no salio exactamente como se esperaba

pasos para que ansible tenga privilegios adminstrativos en el nodo

```
[root@localhost ansible]# echo 'ansible ALL=(ALL) NOPASSWD: ALL' > /etc/sudoers.d/ansible
[root@localhost ansible]# chmod 0440 /etc/sudoers.d/ansible
[root@localhost ansible]# visudo -cf /etc/sudoers.d/ansible
/etc/sudoers.d/ansible: parsed OK
[root@localhost ansible]#
```

bien, paso la prueba de visudo

ahora si con ansible

```
[root@localhost ansible]# ansible web -b -m ansible.builtin.dnf -a "name=tree state=present"
nodo1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "msg": "Nothing to do",
    "rc": 0,
    "results": []
}
[root@localhost ansible]# ansible web -b -m ansible.builtin.dnf -a "name=tree state=present"
nodo1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "msg": "Nothing to do",
    "rc": 0,
    "results": []
}
[root@localhost ansible]#
```

Ha funcionado

4. Comprobar como un modulo si es idempotente a diferencia de la shell

```
[root@localhost ansible]# ansible web -m ansible.builtin.shell -a "echo 'zona=aves' >>
~/notas.txt"
nodo1 | CHANGED | rc=0 >>

[root@localhost ansible]# ansible web -m ansible.builtin.shell -a "echo 'zona=aves' >>
~/notas.txt"
nodo1 | CHANGED | rc=0 >>

[root@localhost ansible]# ansible web -m ansible.builtin.shell -a "echo 'zona=aves' >>
~/notas.txt"
nodo1 | CHANGED | rc=0 >>

[root@localhost ansible]# ansible web -m ansible.builtin.lineinfile -a
"path=/home/ansible/notas2.txt line='zona=aves' create=true"
nodo1 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "backup": "",
    "changed": true,
    "msg": "line added"
}
[root@localhost ansible]# ansible web -m ansible.builtin.lineinfile -a
"path=/home/ansible/notas2.txt line='zona=aves' create=true"
nodo1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "backup": "",
    "changed": false,
    "msg": ""
}
[root@localhost ansible]# ansible web -m ansible.builtin.lineinfile -a
"path=/home/ansible/notas2.txt line='zona=aves' create=true"
nodo1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "backup": "",
    "changed": false,
    "msg": ""
}
[root@localhost ansible]# ssh ansible@nodo1 'cat ~/notas.txt; echo ---; cat ~/notas2.txt'
zona=aves
zona=aves
zona=aves
---
```

```
zona=aves
[root@localhost ansible]#
```

Efectivamente, un script de shell no es idempotente, a diferencia de un modulo se ansible que si lo es.

5. Archivos, usuarios y servicios

```
[root@localhost ansible]# ansible web -m ansible.builtin.copy -a "src=/root/reporte.txt
dest=/home/ansible/reporte-ansible.txt mode=0640"
nodo1 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "checksum": "de403ee2581c4d47f6621b4c92e9b9e246003cb2",
    "dest": "/home/ansible/reporte-ansible.txt",
    "gid": 8006,
    "group": "ansible",
    "md5sum": "e91e656a1c829943186ebb6e74772bc0",
    "mode": "0640",
    "owner": "ansible",
    "secontext": "unconfined_u:object_r:user_home_t:s0",
    "size": 47,
    "src": "/home/ansible/.ansible/tmp/ansible-tmp-1793325250.9809897-7334-
134445662551160/source",
    "state": "file",
    "uid": 8005
}
[root@localhost ansible]# ansible web -b -m ansible.builtin.file -a "path=/opt/zoologico
state=directory mode=0755"
nodo1 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "gid": 8002,
    "group": "cuidadores",
    "mode": "0755",
    "owner": "root",
    "path": "/opt/zoologico",
    "secontext": "unconfined_u:object_r:usr_t:s0",
    "size": 6,
    "state": "directory",
    "uid": 0
}
[root@localhost ansible]# ansible web -b -m ansible.builtin.user -a "name=koala
comment='Cuidador koalas'"
nodo1 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "append": false,
    "changed": true,
    "comment": "Cuidador koalas",
    "group": 8003,
```

```
    "home": "/home/koala",
    "move_home": false,
    "name": "koala",
    "shell": "/bin/bash",
    "state": "present",
    "uid": 8002
}
[root@localhost ansible]# ansible web -b -m ansible.builtin.user -a "name=koala state=absent
remove=true"
nodo1 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": true,
    "force": false,
    "name": "koala",
    "remove": true,
    "state": "absent"
}
[root@localhost ansible]# ansible web -b -m ansible.builtin.service -a "name=sshd
state=started enabled=true"
nodo1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "enabled": true,
    "name": "sshd",
    "state": "started",
    "status": {
        "AccessSELinuxContext": "system_u:object_r:sshd_unit_file_t:s0",
        "ActiveEnterTimestamp": "Thu 2026-10-29 17:25:24 CST",
        "ActiveEnterTimestampMonotonic": "7012491",
        "ActiveExitTimestampMonotonic": "0",
        "ActiveState": "active",
        "After": "network.target systemd-journald.socket basic.target system.slice sshd-
keygen.target sysinit.target",
        "AllowIsolate": "no",
        "AssertResult": "yes",
        "AssertTimestamp": "Thu 2026-10-29 17:25:24 CST",
        "AssertTimestampMonotonic": "6926429",
        "Before": "shutdown.target multi-user.target",
        "BlockIOAccounting": "no",
        "BlockIOWeight": "[not set]",
        "CPUAccounting": "yes",
        "CPUAffinityFromNUMA": "no",
        "CPUQuotaPerSecUSec": "infinity",
        "CPUQuotaPeriodUSec": "infinity",
        "CPUSchedulingPolicy": "0",
```

```
        "CPUSchedulingPriority": "0",
        "CPUSchedulingResetOnFork": "no",
        "CPUShares": "[not set]",
        "CPUUsageNSec": "3827094000",
        "CPUWeight": "[not set]",
        "CacheDirectoryMode": "0755",
        "CanFreeze": "yes",
        "CanIsolate": "no",
        "CanReload": "yes",
        "CanStart": "yes",
        "CanStop": "yes",
        "CapabilityBoundingSet": "cap_chown cap_dac_override cap_dac_read_search cap_fowner
cap_fsetid cap_kill cap_setgid cap_setuid cap_setpcap cap_linux_immutable
cap_net_bind_service cap_net_broadcast cap_net_admin cap_net_raw cap_ipc_lock cap_ipc_owner
cap_sys_module cap_sys_rawio cap_sys_chroot cap_sys_ptrace cap_sys_pacct cap_sys_admin
cap_sys_boot cap_sys_nice cap_sys_resource cap_sys_time cap_sys_tty_config cap_mknod
cap_lease cap_audit_write cap_audit_control cap_setfcap cap_mac_override cap_mac_admin
cap_syslog cap_wake_alarm cap_block_suspend cap_audit_read cap_perfmon cap_bpf
cap_checkpoint_restore",
        "CleanResult": "success",
        "CollectMode": "inactive",
        "ConditionResult": "yes",
        "ConditionTimestamp": "Thu 2026-10-29 17:25:24 CST",
        "ConditionTimestampMonotonic": "6926420",
        "ConfigurationDirectoryMode": "0755",
        "Conflicts": "shutdown.target",
        "ConsistsOf": "sshd-keygen.target",
        "ControlGroup": "/system.slice/sshd.service",
        "ControlGroupId": "2977",
        "ControlPID": "0",
        "CoredumpFilter": "0x33",
        "DefaultDependencies": "yes",
        "DefaultMemoryLow": "0",
        "DefaultMemoryMin": "0",
        "Delegate": "no",
        "Description": "OpenSSH server daemon",
        "DevicePolicy": "auto",
        "Documentation": "\"man:sshd(8)\" \"man:sshd_config(5)\"",
        "DynamicUser": "no",
        "EnvironmentFiles": "/etc/sysconfig/sshd (ignore_errors=yes)",
        "ExecMainCode": "0",
        "ExecMainExitTimestampMonotonic": "0",
        "ExecMainPID": "858",
        "ExecMainStartTimestamp": "Thu 2026-10-29 17:25:24 CST",
        "ExecMainStartTimestampMonotonic": "6936615",
        "ExecMainStatus": "0",
        "ExecReload": "{ path=/bin/kill ; argv[]=/bin/kill -HUP $MAINPID ; ignore_errors=no ;
start_time=[n/a] ; stop_time=[n/a] ; pid=0 ; code=(null) ; status=0/0 }",
        "ExecReloadEx": "{ path=/bin/kill ; argv[]=/bin/kill -HUP $MAINPID ; flags= ;
```

```
start_time=[n/a] ; stop_time=[n/a] ; pid=0 ; code=(null) ; status=0/0 }",
        "ExecStart": "{ path=/usr/sbin/sshd ; argv[]=/usr/sbin/sshd -D $OPTIONS ;
ignore_errors=no ; start_time=[Thu 2026-10-29 17:25:24 CST] ; stop_time=[n/a] ; pid=858 ;
code=(null) ; status=0/0 }",
        "ExecStartEx": "{ path=/usr/sbin/sshd ; argv[]=/usr/sbin/sshd -D $OPTIONS ; flags= ;
start_time=[Thu 2026-10-29 17:25:24 CST] ; stop_time=[n/a] ; pid=858 ; code=(null) ;
status=0/0 }",
        "ExitType": "main",
        "FailureAction": "none",
        "FileDescriptorStoreMax": "0",
        "FinalKillSignal": "9",
        "FragmentPath": "/usr/lib/systemd/system/sshd.service",
        "FreezerState": "running",
        "GID": "[not set]",
        "GuessMainPID": "yes",
        "IOAccounting": "no",
        "IOReadBytes": "18446744073709551615",
        "IOReadOperations": "18446744073709551615",
        "IOSchedulingClass": "2",
        "IOSchedulingPriority": "4",
        "IOWeight": "[not set]",
        "IOWriteBytes": "18446744073709551615",
        "IOWriteOperations": "18446744073709551615",
        "IPAccounting": "no",
        "IPEgressBytes": "[no data]",
        "IPEgressPackets": "[no data]",
        "IPIngressBytes": "[no data]",
        "IPIngressPackets": "[no data]",
        "Id": "sshd.service",
        "IgnoreOnIsolate": "no",
        "IgnoreSIGPIPE": "yes",
        "InactiveEnterTimestampMonotonic": "0",
        "InactiveExitTimestamp": "Thu 2026-10-29 17:25:24 CST",
        "InactiveExitTimestampMonotonic": "6936908",
        "InvocationID": "4bde42e2c2634d86a64accf58e4ec3bc",
        "JobRunningTimeoutUSec": "infinity",
        "JobTimeoutAction": "none",
        "JobTimeoutUSec": "infinity",
        "KeyringMode": "private",
        "KillMode": "process",
        "KillSignal": "15",
        "LimitAS": "infinity",
        "LimitASSoft": "infinity",
        "LimitCORE": "infinity",
        "LimitCORESoft": "infinity",
        "LimitCPU": "infinity",
        "LimitCPUSoft": "infinity",
        "LimitDATA": "infinity",
        "LimitDATASoft": "infinity",
```

```
        "LimitFSIZE": "infinity",
        "LimitFSIZESoft": "infinity",
        "LimitLOCKS": "infinity",
        "LimitLOCKSSoft": "infinity",
        "LimitMEMLOCK": "8388608",
        "LimitMEMLOCKSoft": "8388608",
        "LimitMSGQUEUE": "819200",
        "LimitMSGQUEUESoft": "819200",
        "LimitNICE": "0",
        "LimitNICESoft": "0",
        "LimitNOFILE": "524288",
        "LimitNOFILESoft": "1024",
        "LimitNPROC": "6647",
        "LimitNPROCSoft": "6647",
        "LimitRSS": "infinity",
        "LimitRSSSoft": "infinity",
        "LimitRTPRIO": "0",
        "LimitRTPRIOSoft": "0",
        "LimitRTTIME": "infinity",
        "LimitRTTIMESoft": "infinity",
        "LimitSIGPENDING": "6647",
        "LimitSIGPENDINGSoft": "6647",
        "LimitSTACK": "infinity",
        "LimitSTACKSoft": "8388608",
        "LoadState": "loaded",
        "LockPersonality": "no",
        "LogLevelMax": "-1",
        "LogRateLimitBurst": "0",
        "LogRateLimitIntervalUSec": "0",
        "LogsDirectoryMode": "0755",
        "MainPID": "858",
        "ManagedOOMMemoryPressure": "auto",
        "ManagedOOMMemoryPressureLimit": "0",
        "ManagedOOMPreference": "none",
        "ManagedOOMSwap": "auto",
        "MemoryAccounting": "yes",
        "MemoryAvailable": "infinity",
        "MemoryCurrent": "9076736",
        "MemoryDenyWriteExecute": "no",
        "MemoryHigh": "infinity",
        "MemoryLimit": "infinity",
        "MemoryLow": "0",
        "MemoryMax": "infinity",
        "MemoryMin": "0",
        "MemoryPeak": "12337152",
        "MemorySwapMax": "infinity",
        "MountAPIVFS": "no",
        "NFileDescriptorStore": "0",
        "NRestarts": "0",
```

```
        "NUMAPolicy": "n/a",
        "Names": "sshd.service",
        "NeedDaemonReload": "no",
        "Nice": "0",
        "NoNewPrivileges": "no",
        "NonBlocking": "no",
        "NotifyAccess": "main",
        "OOMPolicy": "stop",
        "OOMScoreAdjust": "0",
        "OnFailureJobMode": "replace",
        "OnSuccessJobMode": "fail",
        "Perpetual": "no",
        "PrivateDevices": "no",
        "PrivateIPC": "no",
        "PrivateMounts": "no",
        "PrivateNetwork": "no",
        "PrivateTmp": "no",
        "PrivateUsers": "no",
        "ProcSubset": "all",
        "ProtectClock": "no",
        "ProtectControlGroups": "no",
        "ProtectHome": "no",
        "ProtectHostname": "no",
        "ProtectKernelLogs": "no",
        "ProtectKernelModules": "no",
        "ProtectKernelTunables": "no",
        "ProtectProc": "default",
        "ProtectSystem": "no",
        "RefuseManualStart": "no",
        "RefuseManualStop": "no",
        "ReloadResult": "success",
        "ReloadSignal": "1",
        "RemainAfterExit": "no",
        "RemoveIPC": "no",
        "Requires": "system.slice sysinit.target",
        "Restart": "on-failure",
        "RestartKillSignal": "15",
        "RestartUSec": "42s",
        "RestrictNamespaces": "no",
        "RestrictRealtime": "no",
        "RestrictSUIDSGID": "no",
        "Result": "success",
        "RootDirectoryStartOnly": "no",
        "RuntimeDirectoryMode": "0755",
        "RuntimeDirectoryPreserve": "no",
        "RuntimeMaxUSec": "infinity",
        "RuntimeRandomizedExtraUSec": "0",
        "SameProcessGroup": "no",
        "SecureBits": "0",
```

```
        "SendSIGHUP": "no",
        "SendSIGKILL": "yes",
        "Slice": "system.slice",
        "StandardError": "inherit",
        "StandardInput": "null",
        "StandardOutput": "journal",
        "StartLimitAction": "none",
        "StartLimitBurst": "5",
        "StartLimitIntervalUSec": "10s",
        "StartupBlockIOWeight": "[not set]",
        "StartupCPUShares": "[not set]",
        "StartupCPUWeight": "[not set]",
        "StartupIOWeight": "[not set]",
        "StateChangeTimestamp": "Thu 2026-10-29 17:25:24 CST",
        "StateChangeTimestampMonotonic": "7012491",
        "StateDirectoryMode": "0755",
        "StatusErrno": "0",
        "StopWhenUnneeded": "no",
        "SubState": "running",
        "SuccessAction": "none",
        "SyslogFacility": "3",
        "SyslogLevel": "6",
        "SyslogLevelPrefix": "yes",
        "SyslogPriority": "30",
        "SystemCallErrorNumber": "2147483646",
        "TTYReset": "no",
        "TTYVHangup": "no",
        "TTYVTDisallocate": "no",
        "TasksAccounting": "yes",
        "TasksCurrent": "1",
        "TasksMax": "10636",
        "TimeoutAbortUSec": "1min 30s",
        "TimeoutCleanUSec": "infinity",
        "TimeoutStartFailureMode": "terminate",
        "TimeoutStartUSec": "1min 30s",
        "TimeoutStopFailureMode": "terminate",
        "TimeoutStopUSec": "1min 30s",
        "TimerSlackNSec": "50000",
        "Transient": "no",
        "Type": "notify",
        "UID": "[not set]",
        "UMask": "0022",
        "UnitFilePreset": "enabled",
        "UnitFileState": "enabled",
        "UtmpMode": "init",
        "WantedBy": "multi-user.target",
        "Wants": "sshd-keygen.target",
        "WatchdogSignal": "6",
        "WatchdogTimestampMonotonic": "0",
```

```
        "WatchdogUSec": "0"
    }
}
[root@localhost ansible]#
```

6. La documentación vive en la terminal

```
[root@localhost ansible]# ansible-doc -l | wc -l
72
[root@localhost ansible]# ansible-doc ansible.builtin.user
> ANSIBLE.BUILTIN.USER    (/usr/lib/python3.9/site-packages/ansible/modules/user.py)

        Manage user accounts and user attributes. For Windows targets, use the
[ansible.windows.win_user] module instead.

ADDED IN: version 0.2 of ansible-core

OPTIONS (= is mandatory):

- append
        If `true', add the user to the groups specified in `groups'.
        If `false', user will only be added to the groups specified in `groups', removing
them from all other groups.
        default: false
        type: bool

- authorization
        Sets the authorization of the user.
        Does nothing when used with other platforms.
        Can set multiple authorizations using comma separation.
        To delete all authorizations, use `authorization=''.
        Currently supported on Illumos/Solaris.
        default: null
        type: str
        added in: version 2.8 of ansible-core


- comment
        Optionally sets the description (aka `GECOS') of user account.
        default: null
        type: str

- create_home
        Unless set to `false', a home directory will be made for the user when the account is
created or if the home directory does not exist.
        Changed from `createhome' to `create_home' in Ansible 2.5.
        aliases: [createhome]
        default: true
        type: bool

- expires
        An expiry time for the user in epoch, it will be ignored on platforms that do not
support this.
        Currently supported on GNU/Linux, FreeBSD, and DragonFlyBSD.
        Since Ansible 2.6 you can remove the expiry time by specifying a negative value.
```

```
Currently supported on GNU/Linux and FreeBSD.
        default: null
        type: float
        added in: version 1.9 of ansible-core



- force
        This only affects `state=absent', it forces removal of the user and associated
directories on supported platforms.
        The behavior is the same as `userdel --force', check the man page for `userdel' on
your system for details and support.
[root@localhost ansible]# ansible-doc -s ansible.builtin.lineinfile
- name: Manage lines in text files
  lineinfile:
      attributes:           # The attributes the resulting filesystem object should have. To
get supported flags look at the man page for `chattr' on the target system. This string
should contain the attributes
                            # in the same order as the one displayed by `lsattr'. The `='
operator is assumed as default, otherwise `+' or `-' operators need to be included in the
                            # string.
      backrefs:             # Used with `state=present'. If set, `line' can contain
backreferences (both positional and named) that will get populated if the `regexp' matches.
This parameter changes the operati>
                            # of the module slightly; `insertbefore' and `insertafter' will
be ignored, and if the `regexp' does not match anywhere in the file, the file will be
                            # left unchanged. If the `regexp' does match, the last matching
line will be replaced by the expanded line parameter. Mutually exclusive with
                            # `search_string'.
      backup:               # Create a backup file including the timestamp information so
you can get the original file back if you somehow clobbered it incorrectly.
      create:               # Used with `state=present'. If specified, the file will be
created if it does not already exist. By default it will fail if the file is missing.
      firstmatch:           # Used with `insertafter' or `insertbefore'. If set,
`insertafter' and `insertbefore' will work with the first line that matches the given regular
expression.
      group:                # Name of the group that should own the filesystem object, as
would be fed to `chown'. When left unspecified, it uses the current group of the current user
unless you are root, in
                            # which case it can preserve the previous ownership.
      insertafter:          # Used with `state=present'. If specified, the line will be
inserted after the last match of specified regular expression. If the first match is
required, use(firstmatch=yes). A
                            # special value is available; `EOF' for inserting the line at
the end of the file. If specified regular expression has no matches, EOF will be used
                            # instead. If `insertbefore' is set, default value `EOF' will be
ignored. If regular expressions are passed to both `regexp' and `insertafter',
                            # `insertafter' is only honored if no match for `regexp' is
found. May not be used with `backrefs' or `insertbefore'.
      insertbefore:         # Used with `state=present'. If specified, the line will be
```

```
inserted before the last match of specified regular expression. If the first match is
required, use `firstmatch=yes'. A
                              # value is available; `BOF' for inserting the line at the
beginning of the file. If specified regular expression has no matches, the line will be
                              # inserted at the end of the file. If regular expressions are
passed to both `regexp' and `insertbefore', `insertbefore' is only honored if no match for
                              # `regexp' is found. May not be used with `backrefs' or
`insertafter'.
      line:                   # The line to insert/replace into the file. Required for
`state=present'. If `backrefs' is set, may contain backreferences that will get expanded with
the `regexp' capture groups if
                              # the regexp matches.
      mode:                   # The permissions the resulting filesystem object should have.
For those used to `/usr/bin/chmod' remember that modes are actually octal numbers. You must
either add a leading zero so
                              # that Ansible's YAML parser knows it is an octal number (like
`0644' or `01777') or quote it (like `'644'' or `'1777'') so Ansible receives a string and
                              # can do its own conversion from string into number. Giving
Ansible a number without following one of these rules will end up with a decimal number which
                              # will have unexpected results. As of Ansible 1.8, the mode may
be specified as a symbolic mode (for example, `u+rwx' or `u=rw,g=r,o=r'). If `mode' is
                              # not specified and the destination filesystem object *does not*
exist, the default `umask' on the system will be used when setting the mode for the
                              # newly created filesystem object. If `mode' is not specified
and the destination filesystem object *does* exist, the mode of the existing filesystem
                              # object will be used. Specifying `mode' is the best way to
ensure filesystem objects are created with the correct permissions. See CVE-2020-1736 for
                              # further details.
      others:                 # All arguments accepted by the [ansible.builtin.file] module
also work here.
      owner:                  # Name of the user that should own the filesystem object, as
would be fed to `chown'. When left unspecified, it uses the current user unless you are root,
in which case it can preser>
                              # the previous ownership. Specifying a numeric username will be
assumed to be a user ID and not a username. Avoid numeric usernames to avoid this
                              # confusion.
      path:                   # (required) The file to modify. Before Ansible 2.3 this option
was only usable as `dest', `destfile' and `name'.
      regexp:                 # The regular expression to look for in every line of the file.
For `state=present', the pattern to replace if found. Only the last line found will be
replaced. For `state=absent', t>
                              # pattern of the line(s) to remove. If the regular expression is
not matched, the line will be added to the file in keeping with `insertbefore' or
                              # `insertafter' settings. When modifying a line the regexp
should typically match both the initial state of the line as well as its state after
                              # replacement by `line' to ensure idempotence. Uses Python
regular expressions. See https://docs.python.org/3/library/re.html.
      search_string:          # The literal string to look for in every line of the file. This
does not have to match the entire line. For `state=present', the line to replace if the
```

```
string is found in the file.
                              # Only the last line found will be replaced. For `state=absent',
the line(s) to remove if the string is in the line. If the literal expression is not
                              # matched, the line will be added to the file in keeping with
`insertbefore' or `insertafter' settings. Mutually exclusive with `backrefs' and `regexp'.
      selevel:                # The level part of the SELinux filesystem object context. This
is the MLS/MCS attribute, sometimes known as the `range'. When set to `_default', it will use
the `level' portion of t>
                              # policy if available.
[root@localhost ansible]#
```

Bastante información

Parte 3 Playbooks de ansible

Playbook 1 - Usuarios con loop

Creamos el archivo **usuarios.yml** en la carpeta de trabajo /root/devops2026/ansible

```
[root@localhost ansible]# pwd
/root/devops2026/ansible
[root@localhost ansible]# touch usuarios.yml
[root@localhost ansible]# vim usuarios.yml
```

con el siguiente contenido

```
---
- name: Cuidadores del zoológico
  hosts: web
  become: true
  vars:
    cuidadores:
      - koala
      - panda
      - lince
  tasks:
    - name: Crear grupo cuidadores
      ansible.builtin.group:
        name: cuidadores
        state: present

    - name: Crear usuarios
      ansible.builtin.user:
        name: "{{ item }}"
        groups: cuidadores
        append: true
        shell: /bin/bash
      loop: "{{ cuidadores }}"

    - name: Directorio compartido
      ansible.builtin.file:
        path: /opt/zoologico
        state: directory
        group: cuidadores
        mode: "2770"
```

Pasamos a usar el playbook

```
[root@localhost ansible]# ansible-playbook usuarios.yml --syntax-check

playbook: usuarios.yml
[root@localhost ansible]# ansible-playbook usuarios.yml

PLAY [Cuidadores del zoológico]
*****************************************************************************************
*********************************************************************************

TASK [Gathering Facts]
*****************************************************************************************
*****************************************************************************************
***
ok: [nodo1]

TASK [Crear grupo cuidadores]
*****************************************************************************************
*********************************************************************************
ok: [nodo1]

TASK [Crear usuarios]
*****************************************************************************************
*****************************************************************************************
****
changed: [nodo1] => (item=koala)
ok: [nodo1] => (item=panda)
ok: [nodo1] => (item=lince)

TASK [Directorio compartido]
*****************************************************************************************
*********************************************************************************
changed: [nodo1]

PLAY RECAP
*****************************************************************************************
*****************************************************************************************
***************
nodo1                      : ok=4    changed=2    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0

[root@localhost ansible]# ansible-playbook usuarios.yml

PLAY [Cuidadores del zoológico]
*****************************************************************************************
*********************************************************************************

TASK [Gathering Facts]
*****************************************************************************************
```

```
*********************************************************************************************
***
ok: [nodo1]

TASK [Crear grupo cuidadores]
*********************************************************************************************
*********************************************************************************
ok: [nodo1]

TASK [Crear usuarios]
*********************************************************************************************
*********************************************************************************************
****
ok: [nodo1] => (item=koala)
ok: [nodo1] => (item=panda)
ok: [nodo1] => (item=lince)

TASK [Directorio compartido]
*********************************************************************************************
*********************************************************************************
ok: [nodo1]

PLAY RECAP
*********************************************************************************************
*********************************************************************************************
***************
nodo1                      : ok=4    changed=0    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0

[root@localhost ansible]# id panda; ls -ld /opt/zoologico
uid=8003(panda) gid=8004(panda) grupos=8004(panda),8002(cuidadores)
drwxrws---. 2 root cuidadores 6 oct 23 09:40 /opt/zoologico
[root@localhost ansible]#
```

ha funcionado y se ha podido ejecutar dos veces con confianza, ya que es idempotente

Playbook 2 - Servidor web con template handler

Para este don dos archivos **web.yml** y **index.html.j2**

Se crea los archivos en la carpeta de trabajo

```
[root@localhost ansible]# touch web.yml
[root@localhost ansible]# touch index.html.j2
[root@localhost ansible]# vim web.yml
[root@localhost ansible]# vim index.html.j2
```

Con su respectivo contendido

web.yml

```yaml
---
- name: Sitio del zoológico
  hosts: web
  become: true
  vars:
    titulo: "Zoológico DevOps 2026"
  tasks:
    - name: Instalar httpd
      ansible.builtin.dnf:
        name: httpd
        state: present

    - name: Publicar index.html desde plantilla
      ansible.builtin.template:
        src: index.html.j2
        dest: /var/www/html/index.html
        owner: root
        group: root
        mode: "0644"
      notify: Reiniciar httpd

    - name: httpd iniciado y habilitado
      ansible.builtin.service:
        name: httpd
        state: started
        enabled: true

  handlers:
    - name: Reiniciar httpd
      ansible.builtin.service:
        name: httpd
        state: restarted
```

index.html.j2

```
<html>
  <body style="font-family:monospace;background:#111;color:#5af78e">
    <h1>{{ titulo }}</h1>
    <p>Servidor: {{ ansible_facts['hostname'] }}</p>
    <p>SO: {{ ansible_facts['distribution'] }} {{ ansible_facts['distribution_version'] }}
</p>
    <p>Memoria: {{ ansible_facts['memtotal_mb'] }} MB</p>
    <p>Desplegado por Ansible</p>
  </body>
</html>
```

Pasamos a la ejecución del playbook

```
[root@localhost ansible]# ansible-playbook web.yml --check --diff

PLAY [Sitio del zoológico]
*********************************************************************************************
*********************************************************************************************

TASK [Gathering Facts]
*********************************************************************************************
*********************************************************************************************
***
ok: [nodo1]

TASK [Instalar httpd]
*********************************************************************************************
*********************************************************************************************
****
changed: [nodo1]

TASK [Publicar index.html desde plantilla]
*********************************************************************************************
*************************************************************************
--- before
+++ after: /root/.ansible/tmp/ansible-local-1090302izsux4/tmpb_4aa0ah/index.html.j2
@@ -0,0 +1,9 @@
+<html>
+  <body style="font-family:monospace;background:#111;color:#5af78e">
+    <h1>Zoológico DevOps 2026</h1>
+    <p>Servidor: localhost</p>
+    <p>SO: Rocky 9.8</p>
+    <p>Memoria: 1707 MB</p>
+    <p>Desplegado por Ansible</p>
+  </body>
+</html>

changed: [nodo1]

TASK [httpd iniciado y habilitado]
*********************************************************************************************
*************************************************************************
fatal: [nodo1]: FAILED! => {"changed": false, "msg": "Could not find the requested service
httpd: host"}

PLAY RECAP
*********************************************************************************************
*********************************************************************************************
***************
nodo1                      : ok=3    changed=2    unreachable=0    failed=1    skipped=0
rescued=0    ignored=0
```

```
[root@localhost ansible]#
```

Salen colores rojos, pero nada que temer. está comprobando lo que debeŕia hacer la simulación de la ejecucion del playbook

```
+    <p>Memoria: 1707 MB</p>
+    <p>Desplegado por Ansible</p>
+  </body>
+</html>

changed: [nodo1]

TASK [httpd iniciado y habilitado] *********************************************************************************************************
fatal: [nodo1]: FAILED! => {"changed": false, "msg": "Could not find the requested service httpd: host"}

PLAY RECAP *********************************************************************************************************************************
nodo1                      : ok=3    changed=2    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

[root@localhost ansible]# ansible-playbook web.yml

PLAY [Sitio del zoológico] *****************************************************************************************************************
```

La siguiente ejecución sin simulación si será limpia.

```
[root@localhost ansible]# ansible-playbook web.yml

PLAY [Sitio del zoológico]
*********************************************************************************************
*********************************************************************************************

TASK [Gathering Facts]
*********************************************************************************************
*********************************************************************************************
***
ok: [nodo1]

TASK [Instalar httpd]
*********************************************************************************************
*********************************************************************************************
****
changed: [nodo1]

TASK [Publicar index.html desde plantilla]
*********************************************************************************************
*************************************************************************
changed: [nodo1]

TASK [httpd iniciado y habilitado]
*********************************************************************************************
*************************************************************************
changed: [nodo1]

RUNNING HANDLER [Reiniciar httpd]
*********************************************************************************************
*************************************************************************
changed: [nodo1]

PLAY RECAP
*********************************************************************************************
*********************************************************************************************
***************
nodo1                      : ok=5    changed=4    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0

[root@localhost ansible]#
```

curl si nos responde con el html

```
[root@localhost ansible]# curl -s localhost
<html>
  <body style="font-family:monospace;background:#111;color:#5af78e">
    <h1>Zoológico DevOps 2026</h1>
    <p>Servidor: localhost</p>
    <p>SO: Rocky 9.8</p>
    <p>Memoria: 1707 MB</p>
    <p>Desplegado por Ansible</p>
  </body>
</html>
[root@localhost ansible]#
```

Que ansible modifique el html

```
[root@localhost ansible]# ansible-playbook web.yml

PLAY [Sitio del zoológico]
*****************************************************************************************
*****************************************************************************************

TASK [Gathering Facts]
*****************************************************************************************
*****************************************************************************************
***
ok: [nodo1]

TASK [Instalar httpd]
*****************************************************************************************
*****************************************************************************************
****
ok: [nodo1]

TASK [Publicar index.html desde plantilla]
*****************************************************************************************
*************************************************************************
ok: [nodo1]

TASK [httpd iniciado y habilitado]
*****************************************************************************************
*************************************************************************
ok: [nodo1]

PLAY RECAP
*****************************************************************************************
*****************************************************************************************
***************
nodo1                      : ok=4    changed=0    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0

[root@localhost ansible]# ansible-playbook web.yml -e "titulo='Zoológico - turno vespertino'"
--diff

PLAY [Sitio del zoológico]
*****************************************************************************************
*****************************************************************************************

TASK [Gathering Facts]
*****************************************************************************************
*****************************************************************************************
***
ok: [nodo1]
```

```
TASK [Instalar httpd]
*********************************************************************************************
*********************************************************************************************
****
ok: [nodo1]

TASK [Publicar index.html desde plantilla]
*********************************************************************************************
*************************************************************************
--- before: /var/www/html/index.html
+++ after: /root/.ansible/tmp/ansible-local-145735vmdc2r6/tmphi0xewoc/index.html.j2
@@ -1,6 +1,6 @@
 <html>
   <body style="font-family:monospace;background:#111;color:#5af78e">
-    <h1>Zoológico DevOps 2026</h1>
+    <h1>Zoológico - turno vespertino</h1>
     <p>Servidor: localhost</p>
     <p>SO: Rocky 9.8</p>
     <p>Memoria: 1707 MB</p>

changed: [nodo1]

TASK [httpd iniciado y habilitado]
*********************************************************************************************
*************************************************************************
ok: [nodo1]

RUNNING HANDLER [Reiniciar httpd]
*********************************************************************************************
*************************************************************************
changed: [nodo1]

PLAY RECAP
*********************************************************************************************
*********************************************************************************************
***************
nodo1                      : ok=5    changed=2    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0

[root@localhost ansible]#
```

efectivamente lo modifico

```
[root@localhost ansible]#  curl -s localhost
<html>
  <body style="font-family:monospace;background:#111;color:#5af78e">
    <h1>Zoológico - turno vespertino</h1>
    <p>Servidor: localhost</p>
    <p>SO: Rocky 9.8</p>
    <p>Memoria: 1707 MB</p>
    <p>Desplegado por Ansible</p>
  </body>
</html>
[root@localhost ansible]#
```

ahora dice zoologico turno vespertino y no desplegado por ansile

Reto de 10 minutos



se crea el archivo **web-salud.yml** en el directorio de trabajo

```
[root@localhost ansible]# touch web-salud.yml
[root@localhost ansible]# vim web-salud.yml
```

que hace las mismas tareas que **web.yml** original mas la solicitada en el reto

web-salud.yml

```yaml
---
- name: Sitio del zoológico
  hosts: web
  become: true
  vars:
    titulo: "Zoológico DevOps 2026"
  tasks:
    - name: Instalar httpd
      ansible.builtin.dnf:
        name: httpd
        state: present

    - name: Publicar index.html desde plantilla
      ansible.builtin.template:
        src: index.html.j2
        dest: /var/www/html/index.html
        owner: root
        group: root
        mode: "0644"
      notify: Reiniciar httpd

    - name: Crear archivo de salud
      ansible.builtin.copy:
        content: "OK\n"
        dest: /var/www/html/salud.html
        owner: root
        group: root
        mode: "0644"

    - name: httpd iniciado y habilitado
      ansible.builtin.service:
        name: httpd
        state: started
        enabled: true

  handlers:
    - name: Reiniciar httpd
      ansible.builtin.service:
        name: httpd
        state: restarted
```

Ejecutamos este nuevo playbook

```
[root@localhost ansible]# ansible-playbook web-salud.yml

PLAY [Sitio del zoológico]
*****************************************************************************************
*****************************************************************************************

TASK [Gathering Facts]
*****************************************************************************************
*****************************************************************************************
***
ok: [nodo1]

TASK [Instalar httpd]
*****************************************************************************************
*****************************************************************************************
****
ok: [nodo1]

TASK [Publicar index.html desde plantilla]
*****************************************************************************************
*************************************************************************
changed: [nodo1]

TASK [Crear archivo de salud]
*****************************************************************************************
*****************************************************************************
changed: [nodo1]

TASK [httpd iniciado y habilitado]
*****************************************************************************************
*************************************************************************
ok: [nodo1]

RUNNING HANDLER [Reiniciar httpd]
*****************************************************************************************
*************************************************************************
changed: [nodo1]

PLAY RECAP
*****************************************************************************************
*****************************************************************************************
***************
nodo1                      : ok=6    changed=3    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0

[root@localhost ansible]#
```

Y probamos: curl responde [OK]

```
[root@localhost ansible]# curl -s localhost/salud.html
OK
```

y el recap del playbook

```
[root@localhost ansible]# ansible-playbook web-salud.yml | grep -A2 "PLAY RECAP"
PLAY RECAP *********************************************************************
nodo1                      : ok=5    changed=0    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0

[root@localhost ansible]#
```